FUNDAMENTAL THEOREM OF CALCULUS

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Name: Solutions

Use the Fundamental Theorem of Calculus to evaluate the following definite integrals.

1. $\int_1^3 \left(x^2 + 2x - 4\right) \mathrm{d}x$

$$\int_1^3 (x^2+2x-4)\,dx = \int_1^3 x^2\,dx + 2\int_1^3 x\,dx - 4\int_1^3 dx$$
$$= \frac{1}{3}x^3\Big|_1^3 + x^2\Big|_1^3 - 4x\Big|_1^3$$
$$= \frac{1}{3}(3^3-1) + (3^2-1) - 4(3-1)$$
$$= \frac{26}{3} + 8 - 8$$
$$= \boxed{\frac{26}{3}}$$

2. $\int_0^1 \left(1 - 8v^3 + 16v^7\right) \mathrm{d}v$

$$\int_0^1 (1-8v^3+16v^7)\,dx = \int_0^1 dv - 8\int_0^1 v^3\,dv + 16\int_0^1 v^7\,dv$$
$$= v\Big|_0^1 - 8\left(\tfrac{1}{4}\right)v^4\Big|_0^1 + 16\left(\tfrac{1}{8}\right)v^8\Big|_0^1$$
$$= (1-0) - 2(1-0) + 2(1-0)$$
$$= 1 - 2 + 2 = \boxed{1}$$

3. $\int_1^8 x^{-2/3}\,\mathrm{d}x$

$$\int_1^8 x^{-2/3}\,dx = 3x^{1/3}\Big|_1^8$$
$$= 3\left(\sqrt[3]{8} - \sqrt[3]{1}\right)$$
$$= 3(2-1)$$
$$= \boxed{3}$$

4. $\int_{\pi/6}^{\pi/2} \csc(t)\cot(t)\,\mathrm{d}t$

$$\int_{\pi/6}^{\pi/2} \csc(t)\cot(t) = -\csc(t)\Big|_{\pi/6}^{\pi/2}$$
$$= -\left(\csc(\pi/2) - \csc(\pi/6)\right)$$
$$= \csc(\pi/6) - \csc(\pi/2)$$
$$= \frac{1}{\sin(\pi/6)} - \frac{1}{\sin(\pi/2)}$$
$$= \frac{1}{1/2} - 1 = 2 - 1 = \boxed{1}$$

5. $\displaystyle\int_{\pi/4}^{\pi/3} \csc^2(\theta)\,\mathrm{d}\theta$

$$\int_{\pi/4}^{\pi/3} \csc^2(\theta)\,d\theta = -\cot(\theta)\Big|_{\pi/4}^{\pi/3}$$
$$= \cot(\pi/4) - \cot(\pi/3)$$
$$= 1 - \frac{\cos(\pi/3)}{\sin(\pi/3)}$$
$$= 1 - \frac{1/2}{\sqrt{3}/2}$$
$$= 1 - \frac{1}{2}\left(\frac{2}{\sqrt{3}}\right) = \boxed{1 - \frac{\sqrt{3}}{3}}$$

6. $\displaystyle\int_{0}^{\pi/4} \sec(\theta)\tan(\theta)\,\mathrm{d}\theta$

$$\int_{0}^{\pi/4} \sec(\theta)\tan(\theta)\,d\theta = \sec(\theta)\Big|_{0}^{\pi/4}$$
$$= \sec(\pi/4) - \sec(0)$$
$$= \frac{1}{\cos(\pi/4)} - \frac{1}{\cos(0)}$$
$$= \frac{1}{\frac{\sqrt{2}}{2}} - \frac{1}{1}$$
$$= \frac{2}{\sqrt{2}} - 1$$
$$= \frac{2}{\sqrt{2}}\frac{\sqrt{2}}{\sqrt{2}} - 1$$
$$= \frac{2\sqrt{2}}{2} - 1 = \boxed{\sqrt{2} - 1}$$